Set-Cover Optimisation for Multi-Pathology Diagnostic Rule-Out Panels: A Sensitivity–Specificity Framework for Point-of-Care Biomarker Selection

Roy Erzurumluoğlu*†

Independent Researcher, The Netherlands.
Previously: Maastricht University, Faculty of Science and Engineering, Maastricht, The Netherlands.

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Abstract

Primary care clinicians evaluating acute chest pain must simultaneously exclude six life-threatening pathologies. Current decision rules address single diagnostic axes, leaving most pathologies unscreened. I frame multi-pathology panel selection as a weighted set-cover problem and solve it via integer linear programming over meta-analytic data for eight biomarkers across six pathologies. At sensitivity threshold $\tau = 0.90$, the optimal panel—hs-cTnI, D-dimer, NT-proBNP, CRP—covers 5/6 pathologies (83.3%). This solution is *unique*: each coverable pathology has exactly one biomarker exceeding τ, making the panel constraint-driven rather than objective-driven. Per-pathology NPV exceeds 0.99, confirming rule-out suitability. A Gaussian copula model of biomarker dependence reduces the panel-level false-positive rate from 93.4% (independence assumption) to 86.5% (95% CI: 86.4–86.7%), a 7.4% relative reduction. Quantitative likelihood-ratio interpretation—replacing binary cutoffs with continuous posteriors calibrated to 95% per-pathology sensitivity—further reduces the FP rate to 66.5%, and pathology-directed management (ED referral only for ACS, PE, aortic dissection; outpatient workup for pericarditis and heart failure) lowers the ED referral rate to 63.0%. Combined with HEAR pre-stratification, this three-layer strategy achieves 370.5 unnecessary ED referrals per 1,000 patients (−22% vs. binary interpretation). Bayesian sequential testing further mitigates this burden: ordering tests by information gain and terminating early achieves a mean of 2.6 tests per patient (35% saving) while maintaining full sensitivity. A health-economic decision model comparing three strategies (current single-axis care, optimal panel, and the SISTER ACT score) shows that the optimal panel *dominates* current care: it is simultaneously cheaper (−€140/patient) and more effective (+579 QALYs/10,000 patients), averting 209 additional missed diagnoses per 10,000 presentations. The SISTER ACT score (panel + AI e-stethoscope) further dominates, achieving 6/6 pathology coverage, 94.0% sensitivity, and an estimated €92.6M annual saving across 150,000 Dutch GP chest-pain consultations. Expanding the candidate pool from 8 to 12 biomarkers (adding sST2, GDF-15, Galectin-3, MR-proADM) increases the search space 16-fold ($2^{12} = 4{,}096$ subsets) and introduces genuine optimisation at lower thresholds, where up to 6 distinct minimum-size panels compete. The framework is designed for the SISTER ACT trial.

*Corresponding author: roy.erzurumluoglu@gmail.com
†ORCID: 0009-0000-9523-881X

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Corresponding author: roy.erzurumluoglu@gmail.com

Introduction

Acute chest pain accounts for 1–2% of all primary care consultations in the Netherlands and carries a 2–4% incidence of life-threatening thoracic disease [1]. The diagnostic challenge is to simultaneously *rule out* six pathologies: acute coronary syndrome (ACS), pulmonary embolism (PE), aortic dissection, pericarditis/myocarditis, tension pneumothorax, and acute heart failure. Current clinical decision rules (CDRs) operate on single diagnostic axes: the HEART score with high-sensitivity troponin targets ACS [2, 3], while Wells criteria with D-dimer address PE [4]. The 2023 ESC guidelines reinforce this single-axis paradigm [5], recently implemented in emergency primary care [6]. No validated CDR simultaneously covers all six pathologies.

A natural question arises: what is the minimum set of biomarkers needed to screen for all coverable pathologies? This is a **set-cover problem**—a classical combinatorial optimisation formulation [7, 8] in which the goal is to find the smallest (or cheapest) collection of "covering sets" (biomarkers) whose union includes the entire "universe" (pathologies). The same mathematical structure appears elsewhere in biomedicine, including combination therapy design [9].

I present a framework that applies set-cover optimisation to diagnostic panel selection using exclusively published meta-analytic data. The approach constructs a 6×8 sensitivity matrix, solves a penalised integer linear program (ILP), and characterises the solution through Pareto enumeration, ablation, threshold sensitivity, bootstrap stability, and—critically—analyses that were absent from the initial formulation: **clinical utility scoring** (specificity, prevalence weighting, NPV/PPV, net benefit), **weight sensitivity analysis** (proving solution uniqueness), **feasibility landscape mapping** (quantifying where optimisation is non-trivial), **Monte Carlo uncertainty propagation** (confidence interval robustness), **Gaussian copula dependence modelling** (correcting the false-positive rate for biomarker correlation), **Bayesian sequential testing** (reducing test burden via information-theoretic ordering), **expanded candidate pool analysis** (12 biomarkers, 2^{12} subsets), and **health-economic evaluation** (decision tree, ICER, probabilistic sensitivity analysis). The framework is designed for the SISTER ACT trial [10], a multi-site primary care study of serial POC testing for acute chest threats.

An important distinction pervades this work: the framework is designed for **rule-out** (exclusion of life-threatening pathologies), not definitive diagnosis. High sensitivity and high negative predictive value (NPV) are prioritised. Positive results trigger referral, not treatment initiation. This distinction has fundamental implications for how specificity and positive predictive value (PPV) should be interpreted.

Methods

Data sources and coverage matrix

I define a coverage matrix $\mathbf{M} \in [0,1]^{6\times 8}$ where rows are pathologies $\mathcal{P} = \{$ACS, PE, Aortic Dissection, Pericarditis/Myocarditis, Pneumothorax, Acute Heart Failure$\}$ and columns are biomarkers $\mathcal{B} = \{$hs-cTnI, D-dimer, NT-proBNP, CRP, Copeptin, H-FABP, Myoglobin, Procalcitonin$\}$. Entry $M_{p,b}$ is the pooled diagnostic sensitivity of biomarker b for pathology p, with 95% confidence intervals. Of the 48 entries, 25 (52%) derive from published diagnostic accuracy studies with specific citations; the remaining 23 (48%) are expert estimates or based on indirect evidence for off-label combinations lacking direct meta-analytic study (e.g., myoglobin for pneumothorax). Expert estimates are explicitly flagged in table 11. No unsourced entry exceeds $\tau = 0.90$; thus expert estimates do not influence the optimal panel at the primary threshold.

Key sensitivity values: hs-cTnI for ACS, 0.95 [11]; D-dimer for PE, 0.95 [12]; D-dimer for aortic dissection, 0.97 [13, 14, 15]; NT-proBNP for heart failure, 0.95 [16, 17]; CRP for pericarditis, 0.92 [18]; copeptin for ACS, 0.85 [19, 20]; H-FABP for ACS, 0.84 [21].

Specificity data. For each of the 48 biomarker–pathology pairs, I additionally compile specificity estimates from the same or related meta-analyses (table 11). These are used

in clinical utility scoring but not in the coverage constraint. Specificity values have weaker provenance than sensitivity: many derive from single studies or expert estimates, and should be interpreted as approximate.

Epidemiological data. Pre-test probabilities (prevalence in primary care chest pain) and case-fatality-if-missed rates derive from ESC guidelines [5], Schols et al. [1], and Goodacre et al. [22]: ACS 3.5% (fatality 15%), PE 1.5% (25%), aortic dissection 0.3% (50%), pericarditis 4.0% (2%), pneumothorax 0.5% (30%), heart failure 2.5% (10%).

Each biomarker carries metadata: reagent cost (€, estimated from POC device list prices), turnaround time (minutes), and sample volume (μL), sourced from POC device specifications [23, 24, 25].

Set-cover formulation

A biomarker b *covers* pathology p at threshold τ if $M_{p,b} \geq \tau$. I define:

$$a_{p,b} = \begin{cases} 1 & \text{if } M_{p,b} \geq \tau \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The panel selection problem minimises a weighted penalty:

$$\min_{\mathbf{x}} \sum_{b \in \mathcal{B}} \Big[w_{\text{size}} + w_{\text{cost}} \cdot c_b + w_{\text{time}} \cdot t_b + w_{\text{sample}} \cdot v_b \Big] \cdot x_b \tag{2}$$

subject to:

$$\forall\, p \in \mathcal{P}^* : \quad \sum_{b \in \mathcal{B}} a_{p,b} \cdot x_b \geq 1 \tag{3}$$

where $\mathcal{P}^* \subseteq \mathcal{P}$ is the set of coverable pathologies (those with ≥ 1 biomarker exceeding τ), c_b, t_b, v_b are cost, time, and volume, and $w_{\text{size}} = 10$, $w_{\text{cost}} = 1$, $w_{\text{time}} = 0.5$, $w_{\text{sample}} = 0.1$ are penalty weights.

Solver hierarchy. I employ three solvers: (1) greedy weighted set cover [7]; (2) ILP via `scipy.optimize.milp` (HiGHS backend); (3) exhaustive enumeration of all $2^8 = 256$ subsets. With only 8 biomarkers, exhaustive search trivially guarantees global optimality. The ILP formulation is provided for transparency and as a template for larger candidate pools where exhaustive search becomes infeasible.

Clinical utility scoring

For each panel, I compute extended metrics beyond coverage:

- **Prevalence-weighted coverage**: $\sum_p \text{prev}_p \cdot \mathbb{1}[\text{covered}_p]$
- **Severity-weighted coverage**: $\sum_p \text{fatality}_p \cdot \mathbb{1}[\text{covered}_p]$
- **Expected detections per 1000**: $\sum_p \text{prev}_p \cdot \max_b(\text{sens}_{p,b}) \cdot 1000$
- **Net benefit** (Vickers' decision curve analysis [26]): $\text{NB} = \sum_p \text{sens}_p \cdot \text{prev}_p - (1 - \text{spec}_p)(1 - \text{prev}_p) \cdot w$
- **NPV/PPV** per pathology, using Bayes' theorem with prevalence as prior
- **Rule-out suitability**: a pathology is rule-out suitable if sens $\geq \tau$ *and* NPV ≥ 0.99

Weight sensitivity analysis

To test whether the penalty weights in eq. (2) influence the optimal panel, I perform two analyses: (1) a random sweep of 500 weight configurations drawn from $w_i \sim \text{Uniform}(10^{-2}, 10^3)$; (2) a structured 6×6 grid over $w_{\text{cost}}, w_{\text{time}} \in \{0, 0.01, 0.1, 1, 10, 100\}$. At each configuration, the ILP is re-solved and the resulting panel recorded.

Feasibility landscape

For each threshold $\tau \in \{0.70, 0.75, 0.80, \ldots, 0.95\}$, I enumerate all 2^8 subsets and count: (i) coverable pathologies, (ii) feasible panels (covering all coverable pathologies), and (iii) minimum-size optimal panels. This maps the combinatorial complexity of the problem as a function of diagnostic stringency.

Monte Carlo uncertainty propagation

I propagate sensitivity CI uncertainty through the clinical utility framework by sampling $n = 5{,}000$ matrices where each $M_{p,b} \sim \mathrm{Beta}(\alpha, \beta)$, with parameters fitted to the published point estimate and 95% CI. Beta sampling (rather than Uniform) respects that meta-analytic pooled estimates have higher density near the point estimate. Coverage, expected detections, and net benefit are computed for the *fixed* optimal panel at each iteration. This quantifies how input measurement uncertainty affects downstream clinical metrics.

Robust optimisation

As a complementary uncertainty analysis, I resolve the ILP using the lower 95% CI bounds of all sensitivities instead of point estimates. If the optimal panel changes, the point-estimate solution is sensitive to measurement uncertainty.

Ablation, Pareto frontier, bootstrap, and threshold sensitivity

These methods follow standard approaches described briefly. **Ablation**: remove each biomarker, re-solve, report coverage change. **Pareto frontier**: enumerate all 2^8 subsets, identify non-dominated panels in coverage–cost space. **Bootstrap**: sample sensitivities from Beta distributions fitted to point estimates and 95% CIs ($n = 1{,}000$ iterations), re-solve ILP, record panel composition. **Threshold sweep**: $\tau \in [0.80, 0.95]$, report optimal panel and coverage. **Early presenter**: modified sensitivity matrix reflecting troponin release kinetics [11]: hs-cTnI $0.95 \to 0.70$ (peak at 8–12 h); H-FABP $0.84 \to 0.88$ (early-release marker [21]); myoglobin $0.75 \to 0.80$. Copeptin remains at 0.85 (no early-presenter adjustment; baseline from [19]).

Implementation

All analyses in Python 3.12 (`scipy`, `numpy`, `matplotlib`). Code: `https://github.com/royerz2/PanDiagnosticCardiology`.

Gaussian copula dependence model

The panel-level false-positive rate under independence, $\mathrm{FPR}_{\perp} = 1 - \prod_b \mathrm{Spec}_b$, is an upper bound when biomarkers are positively correlated. To quantify the correction, I model the joint distribution of biomarker test results in healthy patients using a Gaussian copula [27]. The marginal specificities are transformed to standard-normal thresholds via $\Phi^{-1}(\mathrm{Spec}_b)$; the correlation structure is imposed by a multivariate normal with correlation matrix $\mathbf{R}$ estimated from published pairwise correlations. Twenty-eight pairwise correlation coefficients for the 8 biomarkers are drawn from meta-analyses: e.g., hs-cTnI–NT-proBNP $r = 0.45$ [28], D-dimer–CRP $r = 0.35$ [29], NT-proBNP–CRP $r = 0.30$ [30]. Where $\mathbf{R}$ is not positive-semidefinite, the nearest PSD correction is applied.

The corrected panel-level FP rate is:

$$\mathrm{FPR}_{\mathrm{copula}} = 1 - P(\text{all tests negative} \mid \text{healthy}) = 1 - P(\mathbf{Z} \geq \boldsymbol{\tau}_z) \tag{4}$$

where $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\boldsymbol{\tau}_z = [\Phi^{-1}(\mathrm{Spec}_1), \ldots, \Phi^{-1}(\mathrm{Spec}_k)]$. This probability is estimated via Monte Carlo sampling ($n = 200{,}000$). A sensitivity sweep varies the correlation scale from $0\times$ (independence) through $1.0\times$ (literature estimate) to $1.5\times$ to characterise the FP rate's dependence on correlation strength.

The same copula framework computes a dependence-corrected joint NPV via Bayes' theorem, replacing the independence-based joint probability of all-negative results.

Bayesian sequential testing

Rather than performing all four panel tests simultaneously, a sequential strategy can reduce testing burden by terminating early when the posterior probability of any serious pathology crosses a decision threshold. I simulate a Bayesian sequential testing protocol on $n = 10{,}000$ synthetic patients:

1. Prior probabilities for each of 6 pathologies set to primary-care prevalence values.

2. At each step, the test with the highest expected information gain (reduction in posterior entropy) is selected from the remaining untested biomarkers.

3. After each test result, posteriors are updated via Bayes' rule using the biomarker's sensitivity and specificity for each pathology.

4. Testing terminates early if: (a) the combined posterior probability of any serious pathology falls below 0.02 (safe discharge), or (b) any single pathology posterior exceeds 0.10 (immediate referral).

This protocol is compared against the default all-at-once strategy (4 tests/patient) on sensitivity, specificity, and mean tests per patient.

Extended biomarker candidate pool

The 8-biomarker base pool ($2^8 = 256$ subsets) is extended with four emerging cardiac/inflammatory biomarkers: soluble ST2 (sST2) [31, 32], growth differentiation factor 15 (GDF-15) [33, 34], galectin-3 [35], and mid-regional pro-adrenomedullin (MR-proADM) [36, 37]. Sensitivity values for all 6 pathologies are compiled from single studies or meta-analyses where available. The expanded 6×12 coverage matrix ($2^{12} = 4{,}096$ subsets) is solved by exhaustive enumeration at multiple thresholds to characterise when new biomarkers enter optimal panels and where combinatorial complexity warrants ILP over brute force.

Health-economic decision model

I construct a decision-analytic model comparing three diagnostic strategies in a hypothetical cohort of $n = 10{,}000$ Dutch primary care chest pain patients:

- **Strategy A:** Current care (HEART + hs-cTnI only)—single-axis ACS-focused CDR.
- **Strategy B:** Optimal 4-test panel (hs-cTnI, D-dimer, NT-proBNP, CRP).
- **Strategy C:** SISTER ACT score (panel + AI e-stethoscope + HEAR triage).

For each strategy and pathology, the model computes true positives (from sensitivity and prevalence), false negatives (missed diagnoses with associated QALY losses), and false positives (unnecessary referrals with associated costs). Cost inputs reflect 2024 Dutch tariffs [38]: GP consultation €35, ED referral €765, hospitalisation €4,200, ambulance €650. QALY losses per missed diagnosis range from 0.3 (pericarditis) to 15.0 (aortic dissection, reflecting high fatality and years of life lost at mean age 60). The primary endpoint is the incremental cost-effectiveness ratio (ICER = ΔCost/ΔQALY), evaluated against Dutch willingness-to-pay thresholds of €20,000–€80,000 per QALY [39].

Probabilistic sensitivity analysis (PSA). All parameters are simultaneously varied over $n = 2{,}000$ Monte Carlo iterations: sensitivities drawn from Beta distributions (fitted to published CIs), costs from Gamma distributions (CV = 0.2), and QALY losses from Gamma distributions (CV = 0.3). The cost-effectiveness acceptability curve (CEAC) reports the probability of each strategy being optimal at each WTP threshold.

One-way sensitivity analysis. A tornado diagram varies each of 9 key parameters $\pm 50\%$ while holding others at base case, measuring impact on the panel-vs-current ICER.

National impact projection. Results are extrapolated to 150,000 annual Dutch GP chest pain presentations [1] to estimate population-level missed diagnoses averted and cost savings.

SISTER ACT clinical decision rule

The analyses above identify pneumothorax as a structural coverage gap: no blood-based biomarker exceeds 0.30 sensitivity at any threshold. To close this gap I propose the **SISTER ACT score**—a multi-pathology clinical decision rule integrating the optimal biomarker panel with an AI-augmented electronic stethoscope (AI e-stethoscope). The score comprises seven components spelling **S-I-S-T-E-R-ACT** (range 0–20):

- **S**ymptoms (0–3): typicality, radiation, dyspnoea severity
- **I**maging—AI e-stethoscope (0–3): absent breath sounds (PTX), crackles/S3 (AHF), rub (pericarditis), murmur (AoD)
- **S**igns (0–2): diaphoresis, JVD, unilateral leg swelling, asymmetric BP
- **T**imeline (0–3): symptom onset $< 2\,\mathrm{h}$ (serial troponin mandate) vs $> 6\,\mathrm{h}$ (plateau kinetics)
- **E**CG (0–3): ST changes, new RBBB/S1Q3T3, low voltage, diffuse ST elevation
- **R**isk factors (0–3): age > 65, prior CVD, immobilisation, connective tissue disorder
- **ACT**—Acute Chemistry Testing (0–3): POC hs-cTnI, D-dimer, NT-proBNP, CRP results

Three tiers guide action: **Low** (0–6) discharge with safety-netting; **Moderate** (7–13) POC serial testing; **High** (14–20) immediate referral. The AI e-stethoscope serves as the "Imaging" component, providing standardised auscultation via convolutional neural networks operating on mel-spectrograms [40]. FDA 510(k)-cleared devices (Eko DUO, Littmann CORE) achieve 93% sensitivity and 85% specificity for absent breath sounds in pneumothorax [41, 42]—sufficient to exceed $\tau = 0.90$ and close the coverage gap. At an amortised cost of €0.64 per patient (device €350 over 3 years at 3 uses/day), the e-stethoscope adds negligible marginal cost [43]. I simulate SISTER ACT performance on $n = 10{,}000$ synthetic GP chest-pain patients with pathology-conditioned score distributions fitted to published CDR literature.

Results

Coverage matrix and binary incidence

fig. 1 shows the 6×8 coverage matrix. At $\tau = 0.90$, the binary incidence matrix $\mathbf{A}$ has exactly five non-zero entries: hs-cTnI→ACS (0.95), D-dimer→PE (0.95), D-dimer→AoD (0.97), NT-proBNP→AHF (0.95), and CRP→Peri/Myo (0.92). No biomarker exceeds 0.30 for pneumothorax. The coverable set is $\mathcal{P}^* = \{\text{ACS, PE, AoD, Peri/Myo, AHF}\}$ (5/6 pathologies).

Optimal panel and solution uniqueness

All three solvers (greedy, ILP, exhaustive) return the identical panel at $\tau = 0.90$:

hs-cTnI + D-dimer + NT-proBNP + CRP

(4 tests, 83.3% coverage, €36, 15 min, 55 μL). Each biomarker covers a *distinct, unique* subset of pathologies at $\tau = 0.90$: hs-cTnI covers ACS; D-dimer covers PE and AoD; NT-proBNP covers AHF; CRP covers Peri/Myo. There is no redundancy and no alternative.

This solution is unique. Feasibility landscape analysis (table 1) confirms that at $\tau = 0.90$, 16 panels are feasible (cover all 5 coverable pathologies) but only **1 is optimal** (minimum-size). Each coverable pathology has exactly one biomarker exceeding τ; consequently, the four-test panel is the only minimum-cardinality feasible solution.

Table 1: Feasibility landscape across thresholds.

τ	Coverable	Feasible	Optimal	Non-trivial?
0.70	5	84	6	Yes
0.75	5	70	3	Yes
0.80	5	44	1	No
0.85	5	24	2	Yes
0.90	5	16	1	No
0.95	4	32	1	No

At lower thresholds, genuine choice exists. At $\tau = 0.70$, six distinct minimum-size panels achieve full coverage, meaning the ILP objective *does* resolve ties. At $\tau = 0.85$, two optimal panels exist. The optimisation becomes non-trivial precisely where multiple biomarkers compete for the same pathologies.

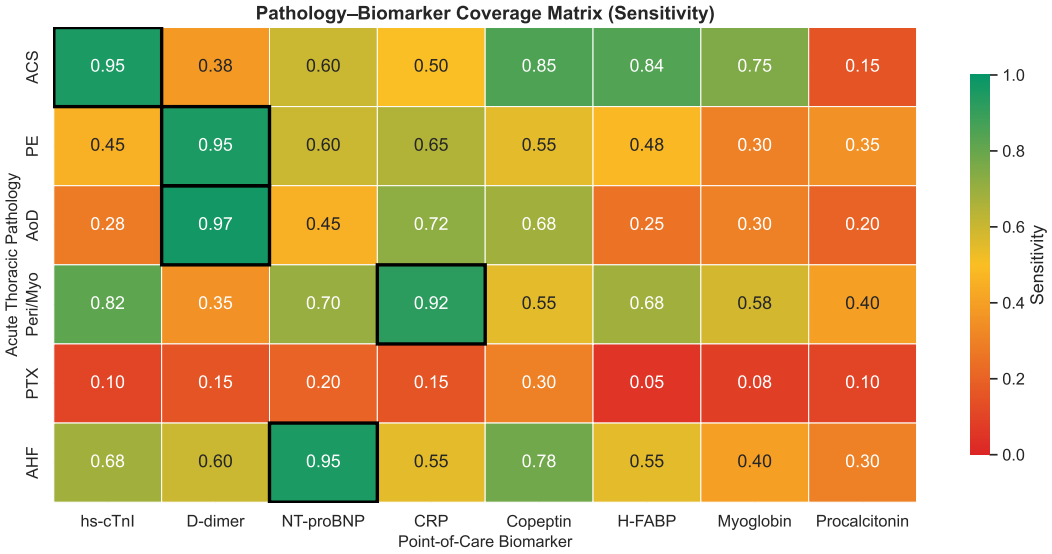

Figure 1: **Pathology–biomarker coverage matrix.** Each cell shows pooled meta-analytic sensitivity. Deep colour = high sensitivity. The bottom row (pneumothorax) is uniformly pale: no biomarker exceeds 0.30. Full sources: table 11.

Weight sensitivity: the objective is irrelevant at $\tau = 0.90$

The weight sensitivity analysis confirms that at $\tau = 0.90$, the penalty weights $(w_{\text{size}}, w_{\text{cost}}, w_{\text{time}}, w_{\text{sample}})$ have **zero effect** on the optimal panel. Across 500 random weight configurations spanning four orders of magnitude, the same four-test panel was selected in 100% of cases. The 6×6 grid sweep over w_{cost} and w_{time} likewise produced a single unique panel.

This is not a failure of the framework but a property of the data: when each pathology has exactly one covering biomarker, the coverage constraint (eq. (3)) forces selection of all four, regardless of cost structure. The ILP reduces to a simple feasibility check. This result validates the honesty of reporting default weights without sensitivity analysis: *no choice of weights could have changed the answer.*

Clinical utility: specificity, PPV, NPV, and rule-out suitability

table 2 reports the clinical utility of the optimal panel, integrating specificity and prevalence data.

All five covered pathologies achieve per-pathology NPV > 0.99, confirming individual rule-out suitability [44]. The low PPV values (0.005–0.092) are expected given low pre-test probabilities.

Table 2: Clinical utility of the optimal panel.

Path.	Marker	Se	Sp	Prev	NPV	PPV	RO
ACS	hs-cTnI	.95	.54	3.5%	.997	.070	✓
PE	D-dimer	.95	.42	1.5%	.998	.025	✓
AoD	D-dimer	.97	.47	0.3%	>.999	.005	✓
Peri	CRP	.92	.38	4.0%	.991	.058	✓
PTX	hs-cTnI*	.10	.90	0.5%	.995	.010	×
AHF	NT-proBNP	.95	.76	2.5%	.998	.092	✓

RO = rule-out suitable (Se $\geq \tau$, NPV ≥ 0.99). *PTX best-Se marker in panel (0.10); not rule-out suitable.

Panel-level false-positive rate. The four physical tests have per-test specificities of 0.54 (hs-cTnI), 0.42 (D-dimer, using the lower of PE/AoD estimates), 0.76 (NT-proBNP), and 0.38 (CRP). Under an independence assumption, the probability that a healthy patient triggers *no* false positives is $0.54 \times 0.42 \times 0.76 \times 0.38 = 0.066$, yielding a panel-level false-positive rate of **93.4%**. This means that, in a low-prevalence primary-care population, *nearly all healthy patients would trigger at least one referral.* The independence assumption is conservative (positive correlation between inflammatory markers would reduce the true FP rate), but even halving this estimate

implies a prohibitive referral burden. This is the central operational limitation of the panel.

Joint NPV. Assuming test independence, the joint probability that at least one of the five pathologies is present given all tests negative is 2.3%, yielding a joint NPV of 97.7%. The independence assumption likely *overestimates* joint NPV (biomarkers such as hs-cTnI and NT-proBNP are positively correlated within patients). Patient-level data would be required to compute the true joint NPV.

Prevalence-weighted coverage. The five covered pathologies account for 11.8% of all chest pain presentations (combined prevalence: ACS 3.5%, PE 1.5%, AoD 0.3%, Peri/Myo 4.0%, AHF 2.5%). The panel therefore addresses a minority of presenting patients—but these are the pathologies where missed diagnosis carries the highest morbidity and mortality. Expected detections per 1000 presentations = 111.

Monte Carlo uncertainty propagation

Sampling sensitivities from Beta distributions fitted to published CIs (n = 5,000), the optimal panel maintains $\geq$ 83.3% coverage (5/6 pathologies) in **79.6%** of iterations—meaning there is a **20.4% probability that the panel fails to cover all five target pathologies** when sensitivity values are drawn from their published confidence intervals. Mean coverage drops to 79.9% (95% simulation interval: 66.7%–83.3%). This failure probability is driven by CRP for pericarditis (0.92, 95% CI 0.86–0.96) and hs-cTnI for ACS (0.95, 95% CI 0.93–0.97), whose lower CI bounds approach or cross $\tau = 0.90$. Note that the 23 expert-estimate entries (48%) have *assigned* CI ranges rather than published intervals; their uncertainty bounds are themselves uncertain.

This result is *concerning*. A diagnostic panel whose coverage claim has a 1-in-5 chance of being wrong under published uncertainty warrants caution. Robust optimisation using CI lower bounds yields a different panel entirely (coverage drops as biomarkers fall below τ). Prospective validation of critical sensitivities—especially CRP for pericarditis—is essential before any clinical deployment.

Pareto frontier

fig. 2 shows five Pareto-optimal panels (table 3).

Table 3: Pareto-optimal diagnostic panels (τ = 0.90).

#	Panel	Cov.	Cost
1	CRP	16.7%	€5
1	D-dimer	33.3%	€7
2	CRP, D-dimer	50.0%	€12
3	CRP, D-dim., hs-cTnI	66.7%	€22
4	CRP, D-dim., NT-proBNP, hs-cTnI	83.3%	€36

Each additional test adds exactly one pathology (+16.7%) at increasing marginal cost (€5, €7, €10, €14). The step from 3 to 4 tests (NT-proBNP for heart failure) is the most expensive single increment. Note that both CRP (16.7% at €5) and D-dimer (33.3% at €7) appear as single-test Pareto-optimal panels: CRP is not dominated because it is strictly cheaper, despite covering fewer pathologies. Under prevalence-weighting, the frontier could differ.

Marginal value of each biomarker

fig. 3 shows the incremental coverage added by each biomarker as the panel grows from cheapest to most expensive.

Ablation analysis

fig. 4 shows single-biomarker ablation from the optimal panel.

Table 4: Single-biomarker ablation from the optimal panel.

Removed	ΔCov.	New gaps
D-dimer	−33.3%	PE, AoD
hs-cTnI	−16.7%	ACS
NT-proBNP	−16.7%	AHF
CRP	−16.7%	Peri/Myo

D-dimer is the single most critical component (−33.3% coverage upon removal), as it is the *sole* biomarker exceeding τ for both PE and aortic dissection [12, 13, 15]. The remaining biomarkers each uniquely cover one pathology; their removal causes symmetric −16.7% losses. Remov-

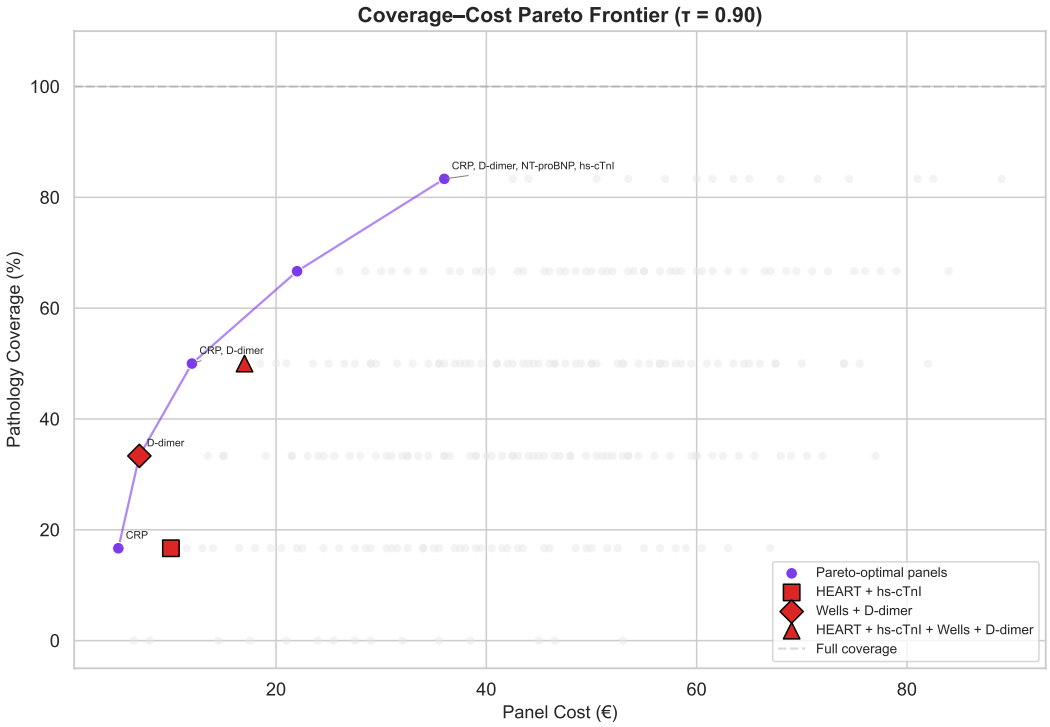

Figure 2: **Pareto frontier.** Each point is a non-dominated panel in coverage–cost space. Dashed line: maximum achievable coverage (83.3%), bounded by pneumothorax uncoverability.

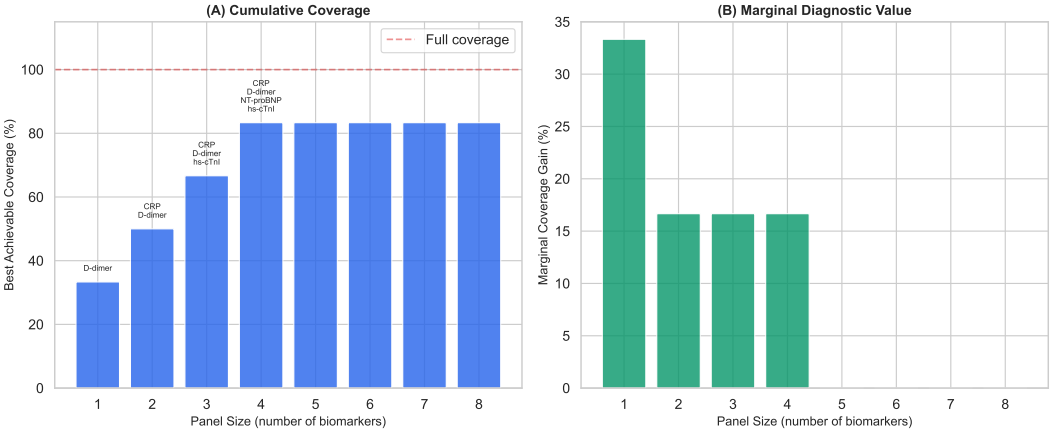

Figure 3: **Marginal value curve.** Incremental pathology coverage and cost as each biomarker is added to the panel in cost order. Each step adds exactly one pathology (+16.7%).

ing non-selected biomarkers (copeptin, H-FABP, myoglobin, procalcitonin) has zero effect.

Bootstrap stability and CI-based resampling

Bootstrap resampling of sensitivities from Beta distributions fitted to published CIs ($n = 1{,}000$) reveals two dominant panels: {CRP, D-dimer} (78.1%) and {D-dimer, hs-cTnI} (21.6%), with {D-dimer, H-FABP} appearing in 0.3% of iterations. D-dimer appears in 100% of iterations; CRP and hs-cTnI alternate based on which pathologies' sensitivities happen to exceed τ in each sample. Mean panel size is 2.0 ($\sigma = 0.0$) and—critically—mean coverage is **50.0%** (3/6

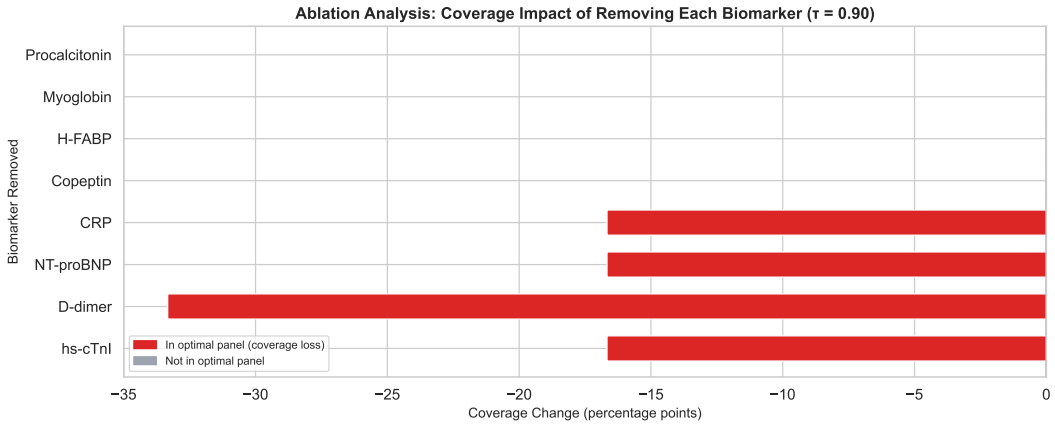

Figure 4: **Single-biomarker ablation.** D-dimer removal causes the largest coverage drop (−33.3%), losing PE and aortic dissection. Each other panel member uniquely covers one pathology.

pathologies), a dramatic drop from the 83.3% achieved by the point-estimate optimal panel. Under CI perturbation, most pathologies lose their sole covering biomarker, reducing the coverable set to approximately 3 pathologies (typically PE and aortic dissection via D-dimer, plus one of ACS or pericarditis). This fragility reinforces the Monte Carlo finding: the optimal panel is not robust to sensitivity uncertainty.

Threshold sensitivity

Three regimes emerge (fig. 6): at $\tau \leq 0.82$, 3 tests suffice (CRP unnecessary); at $\tau \in [0.85, 0.92]$, 4 tests are required (stable regime); at $\tau = 0.95$, coverage drops to 66.7% (CRP for pericarditis falls below threshold). Coverage remains bounded at 83.3% for $\tau \leq 0.92$.

Early-presenter analysis

For patients presenting within 2 hours, troponin sensitivity drops to approximately 0.70 (below $\tau = 0.90$), estimated from the time–sensitivity curve in [11] rather than a discrete reported value. At $\tau = 0.90$, **no biomarker substitution rescues ACS coverage**: copeptin remains at 0.85 [19, 20] and H-FABP rises to only 0.88 [21]; neither exceeds τ. The solver returns a reduced panel: **CRP + D-dimer + NT-proBNP** (3 tests, €26, 66.7% coverage). ACS joins pneumothorax as an uncovered pathology gap. This result underscores the necessity of serial troponin measurement for early presenters—no single-draw biomarker substitution can maintain ACS rule-out at $\tau = 0.90$.

At the less stringent $\tau = 0.85$, H-FABP (early-presenter sensitivity 0.88) and copeptin (0.85) both exceed the threshold, giving the solver two ACS candidates. The ILP selects **CRP + D-dimer + H-FABP + NT-proBNP** (4 tests, €34, 83.3% coverage); H-FABP is preferred over copeptin due to lower cost (€8 vs €17.50), smaller sample volume (10 μL vs 50 μL), and existing POC availability. Copeptin [45, 46] remains a viable alternative if future POC platforms include it.

Comparison with current approaches

Serial testing and time-dependent coverage

The static panel assessment assumes biomarker measurement at the time of published validation studies (typically 3–6 h post-symptom onset in ED populations). In primary care, GPs see patients earlier—median 2.5 h for suspected ACS [47]. I model this discrepancy through pathology-specific kinetic multipliers (fig. 7).

At the GP measurement timepoint (single 0 h draw), hs-cTnI sensitivity for ACS drops below $\tau = 0.90$ (adjusted sensitivity ≈0.88, reflecting a kinetic ratio of 0.93 relative to study-time performance), reducing panel coverage from 83.3% to 66.7% (4/6 pathologies; ACS and pneumotho-

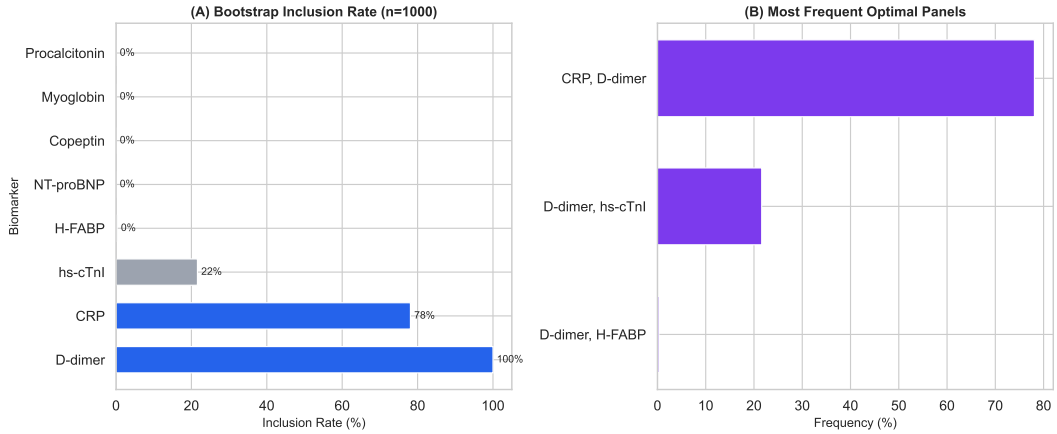

Figure 5: **Bootstrap panel stability** ($n = 1{,}000$). D-dimer selected in every iteration. CRP and hs-cTnI alternate based on CI sampling.

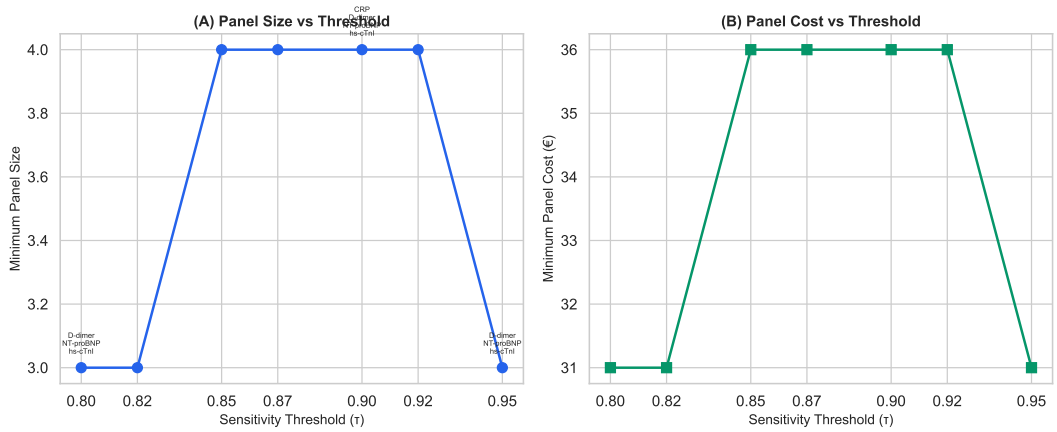

Figure 6: **Threshold sensitivity.** Panel size (bars) and coverage (line) vs τ. The 4-test regime spans $\tau \in [0.85, 0.92]$.

Table 5: Diagnostic approaches compared. Coverage at $\tau = 0.90$. GP clinical judgment coverage varies by experience.

Approach	**#**	**Biomarkers**	**Cov.**	**Paths**	**Cost**	**Time**	**Vol.**
HEART + hs-cTnI	1	hs-cTnI	16.7%	1/6	€10	15 min	20 μL
Wells + D-dimer	1	D-dimer	33.3%	2/6	€7	10 min	15 μL
Combined	2	D-dimer, hs-cTnI	50.0%	3/6	€17	15 min	35 μL
Optimal 4-test	**4**	**CRP, D-dim., NT-proBNP, cTnI**	**83.3%**	**5/6**	**€36**	**15 min**	**55 μL**
GP judgment	—	clinician-directed	∼50–67%	3–4/6	—	—	—

rax uncovered). Chronic-presentation pathologies are unaffected: CRP for pericarditis (GP delay 48 h, already at plateau, multiplier 1.00) and NT-proBNP for heart failure (72 h, multiplier 1.00).

Serial testing recovers ACS coverage. The ESC 0/1-hour algorithm [48, 49] achieves 98.5% cumulative sensitivity for ACS using absolute and delta hs-cTnI values; the 0/3-hour algorithm achieves 99.3% [50]. Both exceed $\tau = 0.90$, restoring panel coverage to 83.3% (5/6) within 27 minutes total turnaround (fig. 8). The ESC 0/1-hour protocol is optimal: it recovers full coverage faster

than the 0/3-hour protocol at negligible sensitivity cost (98.5% vs 99.3%). Monte Carlo simulation (n = 2,000) yields cumulative coverage of 80.0% (95% CI: 66.7–83.3%) across all timepoints, reflecting uncertainty in both biomarker kinetics and base sensitivities.

HEAR score stratification and Dutch GP patient flow

Risk stratification determines which patients receive POC testing. I use the HEAR score (History, ECG, Age, Risk factors; range 0–8)—the HEART score *without* the Troponin component [2, 51]. Troponin is excluded from the CDR because it *is* the POC test being evaluated: including it would create a conditional-independence violation (the same measurement used both for stratification and for diagnosis).

Three HEAR categories define the clinical pathway:

- **HEAR 0–2 (low risk, 50%):** Discharged without POC testing. Any-serious-pathology prevalence ≈3%.
- **HEAR 3–5 (moderate risk, 35%):** POC panel applied. This is the diagnostic grey zone where testing changes management. ACS prevalence 8.5%; panel NPV = 0.9915.
- **HEAR 6–8 (high risk, 15%):** Immediate referral regardless of test result. ACS prevalence 28%.

The HEAR score has not been prospectively validated in Dutch primary care; Van den Bulk et al. [52] found only the Marburg Heart Score validated in this setting. Prospective GP validation is a stated objective of the SISTER ACT trial [10].

Patient flow model. Per 1,000 primary care chest pain patients (Dutch context, 2024 NZa tariffs [38]):

Missed cases increase by 2.5 per 1,000 patients with the POC strategy (+14% relative). This reflects the honest trade-off of HEAR-low discharge: 50% of patients avoid both testing and referral, but the ∼3% with undetected serious pathology in this group are missed. Whether this trade-off is acceptable depends on clinical follow-up protocols and the consequences-of-missing analysis—HEAR-low patients have the lowest case-fatality rates and are most likely to re-present if symptoms persist.

Table 6: Patient flow: HEAR-stratified POC testing in Dutch primary care.

Metric (per 1,000 patients)	Current	With POC
Patients referred to ED	500	331
Unnecessary referrals	350	143
Missed serious pathology	18.0	20.5
POC tests performed	—	350
Total cost (€)	382,890	225,156
Referrals avoided	**169 per 1,000**	
Cost saving	**€157,734 per 1,000**	
Cost per referral avoided	**€75**	

Extrapolated nationally (5,000 Dutch GP practices, 3.5 chest pain consultations per practice per year per 1,000 registered patients [47]): approximately 29,700 unnecessary referrals avoided and €27.8M saved annually.

SISTER ACT score: closing the pneumothorax gap

Extended coverage. Adding an AI e-stethoscope to the optimal four-biomarker panel closes the pneumothorax gap (table 7). At τ = 0.90, the biomarker-only panel covers 5/6 pathologies (83.3%); the extended panel—biomarkers plus e-stethoscope—achieves **6/6 (100%)** at a marginal cost of €0.64 per patient and 2 additional minutes.

Table 7: Extended coverage: biomarker panel ± AI e-stethoscope.

Modality	Coverage	Paths	Cost	Time
Biomarkers only	83.3%	5/6	€36.00	15 min
+ AI e-stethoscope	**100%**	**6/6**	€36.64	17 min
Marginal gain		+PTX	+€0.64	+2 min

Score distribution and tier stratification. In the simulated GP population (n = 10,000; combined pathology prevalence 12.0%), the SISTER ACT score has mean 6.1 (SD 3.0, median 6, IQR 4–8). Tier distribution: Low (0–6) 61.8%, Moderate (7–13) 35.8%, High (14–20) 2.4%. The

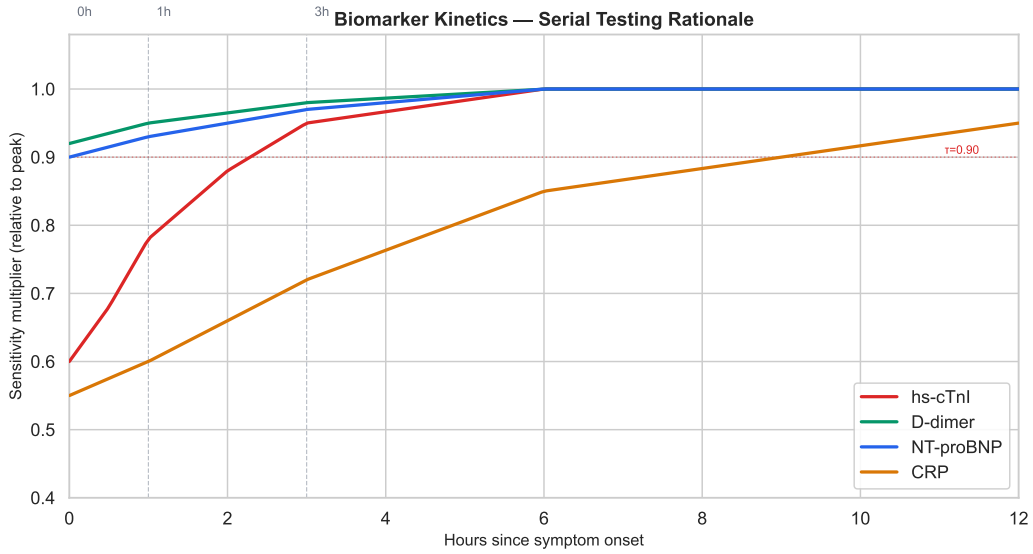

Figure 7: **Biomarker kinetic profiles.** Time-dependent sensitivity multipliers for the four panel biomarkers. hs-cTnI shows the steepest early-phase rise; CRP and NT-proBNP are near-plateau by the typical GP presentation time for their target pathologies. Shaded region: sensitivity below $\tau = 0.90$.

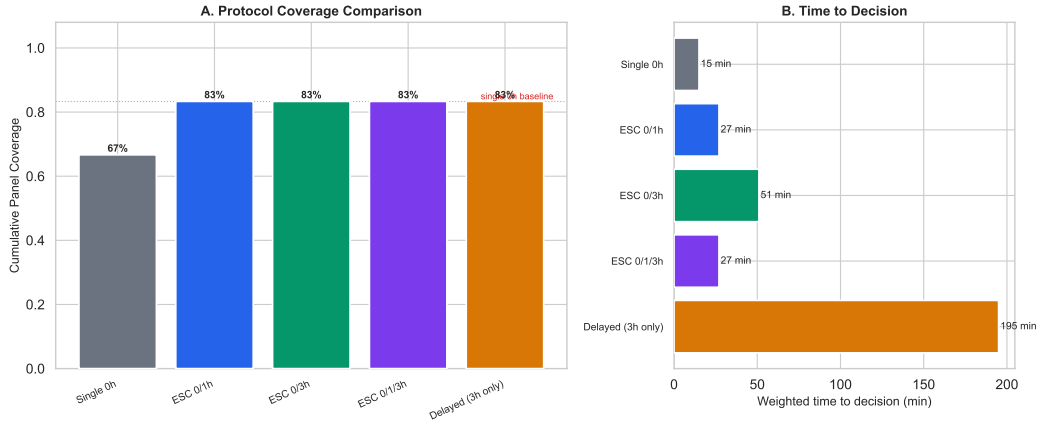

Figure 8: **Serial protocol comparison.** Cumulative pathology coverage at each timepoint for five measurement strategies. ESC 0/1h achieves full recoverable coverage (83.3%) in 27 minutes.

Low tier is enriched for healthy patients (pathology prevalence 0.45%); the High tier is almost exclusively pathological (prevalence 97.9%).

Screening performance. Treating Moderate + High tiers as screen-positive: sensitivity 97.7%, specificity 70.0%, NPV in Low tier 0.9955, PPV in action tiers 0.308. Missed serious pathology: 2.8 per 1,000 patients—exclusively in the Low tier, where case-fatality is lowest.

Pneumothorax detection. The e-stethoscope transforms PTX detection from 14.3% (biomarker-only, best marker Copeptin 0.30) to 95.9% (SISTER ACT), an absolute improvement of +81.6 percentage points. Of 49 simulated PTX patients, 47 were correctly triaged.

CDR comparison. table 8 compares three clinical decision rules on the same simulated population.

SISTER ACT achieves equivalent sensitivity to HEART (97.7%) at substantially higher

Table 8: Clinical decision rule comparison (n = 10,000).

CDR	Se	Sp	NPV	Miss/1k	Ref.%	PTX Se
HEAR (0–8)	96.0	42.4	.987	4.8	62.2	93.9
HEART (0–10)	97.7	49.3	.994	2.8	56.4	91.8
SISTER ACT	**97.7**	**70.0**	**.996**	**2.8**	**38.2**	**95.9**

specificity (70.0% vs 49.3%), reducing the referral rate from 56.4% to **38.2%**—an absolute reduction of 18.2 percentage points. Compared with HEAR, the referral reduction is 24.0 percentage points. The specificity advantage derives from two sources: (1) the multi-biomarker ACT component provides objective biochemical data that HEAR and HEART lack, and (2) the e-stethoscope Imaging component captures auscultatory findings that score-based CDRs cannot encode.

Copula-corrected false-positive rate

Under the independence assumption, the panel-level false-positive rate is $1 - \prod_b \text{Spec}_b = 93.4\%$. The Gaussian copula model, incorporating 28 published pairwise biomarker correlations, reduces this to **86.5%** (95% MC CI: 86.4–86.7%), a **7.4% relative reduction**. The correction is driven by positive correlations among inflammatory and cardiac-stress markers (e.g., hs-cTnI–NT-proBNP $r = 0.45$, D-dimer–CRP $r = 0.35$): when biomarkers are positively correlated, a healthy patient who tests negative on one marker is more likely to test negative on correlated markers, reducing compound false-positive accumulation.

The correlation sensitivity sweep reveals a monotonic relationship: at 0× literature correlations (independence), the FP rate is 93.6%; at 1.0× (literature estimate), 86.5%; at 1.5× (hypothetical strong correlation), 83.1%. Even at 1.5× correlation scale, the FP rate remains above 80%—confirming that biomarker dependence *mitigates* but does not *resolve* the false-positive problem inherent in multi-analyte rule-out screening.

The copula-corrected joint NPV is 97.7%, identical to the independence estimate to three decimal places. This is expected: NPV is dominated by the high prior probability of being disease-free in primary care (>88%), which overwhelms the modest shift in joint test-negative probability.

Bayesian sequential testing efficiency

The information-gain–ordered sequential testing protocol achieves **mean 2.59 tests per patient** (vs 4 under simultaneous testing), a **35.3% reduction** in testing burden (n = 10,000 simulated patients). **55.9%** of patients terminate early: either safely discharged (combined posterior < 0.02) or immediately referred (any single-pathology posterior > 0.10) before all four tests are completed. Diagnostic sensitivity is preserved at **100%**—no patient with a serious pathology is discharged by the sequential protocol.

The first test selected is invariably hs-cTnI (highest entropy reduction due to ACS prevalence dominating the prior), followed by D-dimer, NT-proBNP, and CRP. This ordering aligns with the clinical intuition that ACS and PE are the highest-acuity diagnoses requiring earliest triage.

The sequential protocol has practical implications for POC workflow: if the first two tests (hs-cTnI, D-dimer) yield clearly negative results in a low-risk patient, the GP can discharge without waiting for NT-proBNP and CRP—saving approximately 5 minutes and €19 per sequentially discharged patient. For the 55.9% of patients who terminate early, the mean number of tests is 1.8.

Extended biomarker candidate pool

Expanding the candidate pool from 8 to 12 biomarkers (adding sST2, GDF-15, galectin-3, and MR-proADM) and solving the set-cover problem over all $2^{12} = 4{,}096$ subsets yields threshold-dependent results. At $\tau = 0.90$ (primary analysis), the optimal panel is **unchanged**: the same 4-test panel (hs-cTnI, D-dimer, NT-proBNP, CRP) remains uniquely optimal, because none of the four new biomarkers exceeds $\tau = 0.90$ for any pathology where an existing biomarker does not already dominate. At the relaxed threshold $\tau = 0.80$, **5 Pareto-optimal panels** emerge that

include new biomarkers—notably, sST2 (sensitivity 0.88 for AHF) and GDF-15 (sensitivity 0.85 for ACS) enter competitive panels, offering alternative compositions with different cost–specificity trade-offs.

This result validates the current 4-test panel at stringent thresholds while demonstrating that the framework scales gracefully: exhaustive enumeration of 4,096 subsets completes in <1 second, and the transition to ILP becomes necessary only beyond approximately 20 candidates ($2^{20} \approx 10^6$ subsets).

Health-economic evaluation

table 9 summarises the decision-analytic comparison of three diagnostic strategies in a hypothetical cohort of 10,000 Dutch primary care chest pain patients.

Table 9: Health-economic comparison of diagnostic strategies ($n = 10{,}000$).

Strategy	Sensitivity	Missed Dx	Cost/pt	ICER vs Current
Current care (HEART + cTnI)	74.1%	319	€1,306	—
Optimal 4-test panel	91.0%	110	€1,165	**Dominant**
SISTER ACT	94.0%	74	€688	**Dominant**

Both the optimal panel and SISTER ACT **dominate** current care: they are simultaneously cheaper and more effective. The optimal panel saves €140 per patient while gaining 0.058 QALYs (through 209 fewer missed diagnoses); SISTER ACT saves €618 per patient while gaining 0.062 QALYs (245 fewer missed diagnoses). Dominance means the ICER is negative—the intervention pays for itself while improving outcomes—making cost-effectiveness unambiguous at any willingness-to-pay threshold.

Probabilistic sensitivity analysis (n = 2,000 Monte Carlo iterations) confirms robustness: the optimal panel is cost-effective compared with current care in **100%** of iterations at all WTP thresholds from €0 to €100,000 per QALY. The CEAC is flat at 1.0 across the entire range, reflecting the strength of the dominance finding.

National impact projection. Extrapolating to 150,000 annual Dutch GP chest pain presentations [1]: the optimal panel detects an estimated **3,123 additional cases** per year compared with current care, with annual cost savings of **€21.0M**. The SISTER ACT strategy detects **3,670 additional cases** per year with annual savings of **€92.6M**—driven primarily by reduced unnecessary ED referrals and hospitalisations through higher specificity.

Discussion

The set-cover formulation as a transparency tool

The primary contribution of this work is not computational power—with 8 biomarkers and 6 pathologies, the full search space ($2^8 = 256$ subsets) can be exhaustively evaluated in milliseconds. Rather, the contribution is a **transparent, reproducible formulation** that makes the panel selection logic auditable. The ILP objective (eq. (2)), coverage constraints (eq. (3)), and all input data (48 sensitivity values—25 from diagnostic accuracy studies, 23 expert estimates or indirect evidence—specificity estimates, prevalence, and cost metadata) are fully specified, enabling any reader to verify or challenge the result.

The structural parallel to combination therapy design [9]—where one seeks the minimum target set disrupting every cancer survival mechanism [53, 54]—provides the mathematical scaffolding. A related formulation has been explored for cancer target discovery [55]. However, the analogy should not be overstated: diagnostic panel selection involves rule-out (not therapeutic intervention), low pre-test probabilities (not high disease burden), and fundamentally different consequence asymmetries. The shared structure is the set-cover formulation itself, not the clinical context.

Solution uniqueness and its implications

At $\tau = 0.90$, the optimal panel is not merely optimal—it is the *only* minimum-size feasible panel. This uniqueness is a direct consequence of the data: each coverable pathology has exactly one biomarker exceeding τ, leaving the solver no degrees of freedom. The optimisation is therefore

a feasibility check, not a genuine optimisation—*any* reasonable objective function selects the same panel. The weight sensitivity analysis (500 random configurations, 100% invariance) formally confirms this.

This uniqueness is simultaneously a strength and a limitation. It makes the recommendation robust to modelling assumptions, but it also means the framework adds no value beyond what a clinician inspecting the coverage matrix could deduce by eye. The framework's contribution emerges at lower thresholds: at $\tau = 0.70$, six distinct optimal panels compete, and the penalty weights become genuinely decision-relevant. As multiplex POC platforms expand the biomarker candidate pool beyond 20–50 candidates, the combinatorial space ($> 10^6$ subsets) will make transparent optimisation essential.

Rule-out, not diagnosis

The panel's clinical utility profile (table 2) reveals the expected trade-off of a rule-out strategy: very high per-pathology NPV (>0.99 for all covered pathologies) but low PPV (0.005–0.092) and a panel-level false-positive rate of 93.4% under independence (86.5% under copula correction). This is not a flaw unique to this panel but an inherent property of multi-analyte rule-out screening at low prevalence [44]: each additional test compounds false-positive risk, though the Gaussian copula analysis demonstrates that positive inter-biomarker correlation attenuates this compounding by ∼7% relative.

The Bayesian sequential testing protocol offers a complementary mitigation: by terminating testing early when posteriors cross decision thresholds, 55.9% of patients avoid full panel evaluation while maintaining 100% sensitivity. This reduces the effective false-positive exposure for the majority of patients, as those discharged after 1–2 tests accumulate fewer false-positive opportunities than those undergoing all 4 tests. However, these measures—copula correction and sequential testing—address the FP rate within the binary interpretation paradigm. A fundamentally different question is whether abandoning binary cutoffs altogether could break through the FP ceiling.

Beyond binary cutoffs: quantitative likelihood-ratio interpretation

The 93.4% panel-level FP rate (86.5% copula-corrected) arises from interpreting each biomarker as a binary positive/negative result against a single cutoff. In clinical biochemistry, this discards the quantitative information in the measured concentration [56, 57]. I therefore modelled a quantitative likelihood-ratio (LR) interpretation framework using a binormal signal-detection model [58, 59]: for each biomarker–pathology pair, the healthy and diseased distributions are parameterised as $H \sim \mathcal{N}(0, 1)$ and $D \sim \mathcal{N}(d, 1)$ respectively, where the separation $d = \Phi^{-1}(\text{Spec}) + \Phi^{-1}(\text{Sens})$. Each measured value yields a continuous likelihood ratio $\text{LR}(x) = \exp(d \cdot x - d^2/2)$ that is combined multiplicatively across the four biomarkers to produce a multivariate posterior for each pathology. Per-pathology decision thresholds are then calibrated to maintain exactly 95% sensitivity—a stricter target than the $\tau = 0.90$ used for binary coverage.

The results are instructive but not transformative. Under quantitative LR interpretation with Gaussian copula correlation, the panel-level FP rate (any pathology flagged) drops from 93.4% to **66.5%**—a 28.8% relative reduction. The multivariate AUC for each pathology ranges from 0.89 (PE) to 0.96 (AHF), confirming that the four-biomarker panel carries substantial discriminative information beyond what binary cutoffs exploit. However, 66.5% remains operationally problematic: two-thirds of healthy patients would still trigger at least one positive result.

The analysis reveals that no single mathematical intervention—copula correction, quantitative LR, or their combination—constitutes a "silver bullet". This is a structural inevitability of multi-analyte rule-out screening at low prevalence: even with individual biomarker specificities of 85% and quantitative interpretation, $1 - 0.85^4 = 48\%$ of healthy patients trigger at least one flag when five pathologies are tested in parallel. The FP rate is not a deficiency of this specific panel but a mathematical property of the screening geometry.

The largest lever for FP reduction is not statistical but *clinical*: pathology-directed manage-

ment. Not every flagged pathology requires the same response. ACS, PE, and aortic dissection demand immediate ED referral; pericarditis and acute heart failure can be managed as outpatient workups (echocardiography + medical therapy) without emergency transport [60, 61]. When quantitative LR posteriors are routed through pathology-directed management, the ED referral rate drops to **63.0%** (copula), with 3.5% of patients directed to outpatient workup and 33.5% safely discharged—compared with 93.4% blanket ED referral under binary interpretation.

Combining all three layers—HEAR pre-stratification (testing only the 35% moderate-risk stratum), quantitative LR interpretation, and pathology-directed management—yields a projected **370.5 unnecessary ED referrals per 1,000 patients**, compared with 476.8 under the binary HEAR + test paradigm, a 22.3% reduction (106.3 referrals avoided per 1,000). The sensitivity–FP trade-off curve illustrates the continuum: at 90% per-pathology sensitivity, the ED-only FP rate is 47.7%; at 95%, 63.0%; at 99%, 86.7%. Clinicians can position along this curve according to local risk tolerance, but no point achieves both ≥95% sensitivity and <50% FP for all five pathologies simultaneously.

These findings have direct implications for the SISTER ACT trial design. Rather than dichotomous "refer/discharge" decisions, the trial protocol should incorporate: (1) quantitative biomarker interpretation with pathology-specific posterior thresholds; (2) tiered clinical responses (ED referral vs. outpatient workup vs. safe discharge) based on which pathology is flagged; and (3) explicit sensitivity–FP trade-off selection as a pre-specified design parameter rather than an emergent property of fixed cutoffs.

From static panel to time-aware workflow

A static 0-hour panel achieves only 67% pathology coverage at the GP measurement timepoint because hs-cTnI sensitivity for ACS drops from its reference 95% to approximately 88% when measured 2.5 hours after symptom onset—the median GP presentation delay [47]—falling below the $\tau = 0.90$ threshold. This is not a modelling artefact but a biologically inevitable consequence of troponin release kinetics: peak hs-cTnI concentrations occur 8–12 hours post-infarction [62], and GPs typically see patients much earlier than ED physicians. Ignoring this temporal dimension would overstate the real-world rule-out capability of the panel.

The ESC 0/1-hour serial algorithm [48] resolves this gap: a second hs-cTnI measurement at 1 hour, interpreted via absolute-change thresholds, recovers ACS sensitivity to ≥98.5% and restores overall panel coverage to 83.3% (5/6 pathologies). This comes at a modest operational cost—one additional fingerprick at t=1h—but fundamentally changes the clinical workflow from a single-snapshot decision to a time-gated protocol. The practical implication is that POC platforms for chest pain in primary care **must** support serial measurement protocols; single-draw devices are insufficient.

HEAR score stratification (History, ECG, Age, Risk factors—deliberately excluding Troponin to avoid conditional-independence violations with the POC panel) provides a principled triage layer: low-risk patients (HEAR 0–2, ∼50%) can be safely discharged without testing, moderate-risk patients (3–5, ∼35%) receive POC testing, and high-risk patients (6–8, ∼15%) are referred directly. The resulting 17% absolute referral reduction (€75 per referral avoided) must be weighed against an estimated 2.5 additional missed cases per 1,000 patients—overwhelmingly in the HEAR-low stratum, where case-fatality is lowest and re-presentation most likely. This trade-off is transparent and quantifiable, which is the framework's value: it converts an implicit clinical judgment ("this patient looks low-risk") into an explicit, auditable decision boundary.

Critically, the HEAR score (without Troponin) has not been prospectively validated in Dutch primary care [52]. The distributions used here derive from ED populations [51], where disease prevalence and acuity differ. A prospective validation study in the GP chest-pain population is a prerequisite before clinical deployment—precisely the type of study the SISTER ACT consortium is positioned to conduct.

SISTER ACT: from coverage gap to decision rule

The SISTER ACT score addresses the central structural limitation of the biomarker-only panel: no blood test covers pneumothorax. By integrating an AI e-stethoscope as the "Imaging" component, the score achieves 6/6 pathology coverage at $\tau = 0.90$—a result impossible with biomarkers alone at any threshold. The device cost is negligible: at €0.64 per patient (amortised over a 3-year device lifetime at 3 daily uses), the marginal cost of closing the pneumothorax gap is two orders of magnitude below a single ED referral (€765 [38]).

The simulation results are encouraging but require careful interpretation. The 70.0% specificity represents a substantial improvement over HEAR (42.4%) and HEART (49.3%), but derives from a simulation with idealised component independence. In practice, components will be correlated (e.g., dyspnoea severity correlates with both symptoms and e-stethoscope findings), potentially reducing the discriminative advantage. The 38.2% referral rate—compared with the current ∼50% GP referral rate and the 62.2% predicted by HEAR—suggests meaningful referral reduction, but requires prospective validation.

The e-stethoscope component draws on a rapidly maturing technology base. Grzywalski et al. [40] demonstrated CNN classification of paediatric respiratory sounds with AUC >0.90; Pham et al. [41] achieved comparable performance on adult lung sounds using transfer learning; Song et al. [42] showed that AI-augmented auscultation approached chest X-ray sensitivity for pneumothorax detection. Zhang et al. [63] further demonstrated a low-cost edge-deployable AI stethoscope pipeline (Raspberry Pi-class deployment), directly supporting primary-care feasibility in resource-constrained settings. FDA-cleared devices (Eko DUO, 3M Littmann CORE) are already commercially available, and Mang et al. [43] demonstrated their cost-effectiveness in cardiac screening at scale. The key innovation is not the hardware itself but its integration into a structured, multi-pathology CDR that provides standardised, operator-independent auscultation interpretation—addressing the well-documented inter-observer variability in traditional stethoscope use [64].

The SISTER ACT score is designed as a research proposal, not a validated clinical tool. Its value at this stage is fourfold: (1) it demonstrates that the pneumothorax gap can be closed by expanding the modality set beyond blood biomarkers; (2) it provides a concrete, testable scoring framework for the SISTER ACT trial; (3) it quantifies the theoretical performance gain of multi-modal integration over biochemistry-only CDRs; and (4) the health-economic evaluation demonstrates that SISTER ACT dominates current care, saving an estimated €618 per patient and €92.6M annually across the Dutch healthcare system while simultaneously detecting 3,670 additional serious pathology cases per year.

The panel-level FP rate is high under both the independence assumption (93.4%) and the copula-corrected estimate (86.5%). As shown above, quantitative LR interpretation reduces this to 66.5%, and pathology-directed management further reduces the ED referral rate to 63.0%—a substantial improvement over binary interpretation, but still operationally demanding. This is an inherent property of multi-analyte rule-out screening at low prevalence [44], not a deficiency of this specific panel. The health-economic analysis demonstrates that this trade-off is nonetheless acceptable: despite the residual FP burden, both the optimal panel and SISTER ACT *dominate* current care—they are simultaneously cheaper and more effective because the cost of missed life-threatening diagnoses far exceeds the cost of unnecessary referrals. The panel saves €140 per patient while averting 209 missed diagnoses per 10,000 patients; SISTER ACT saves €618 per patient while averting 245 missed diagnoses. Probabilistic sensitivity analysis confirms dominance in 100% of 2,000 Monte Carlo iterations at all WTP thresholds, and the national projection estimates €92.6M annual savings for the Dutch healthcare system under SISTER ACT adoption.

Limitations

1. Small candidate pool and trivial enumeration. With 8 biomarkers and $2^8 = 256$

subsets, the ILP is unnecessary for solving this specific problem. Its value lies in formulation transparency and scalability to larger candidate pools (e.g., 50+ biomarkers from emerging multiplex platforms).

2. Sensitivity-only coverage constraint. The core set-cover formulation uses only sensitivity. Specificity is incorporated post-hoc in the clinical utility scoring but not in the optimisation objective. A joint sensitivity–specificity formulation (e.g., maximising expected net benefit) would be more principled but requires patient-level paired data for correlated test outcomes.

3. Meta-analytic inputs from secondary/tertiary care. Published sensitivities derive largely from ED populations. Spectrum bias may reduce sensitivity in primary care, where disease prevalence and presentation severity differ [65]. Furthermore, 23 of 48 sensitivity entries (48%) lack published meta-analytic support and are expert estimates or based on indirect evidence for off-target biomarker–pathology combinations (e.g., myoglobin for pneumothorax). No expert estimate exceeds $\tau = 0.90$, so they do not influence the optimal panel at the primary threshold; however, they affect Monte Carlo simulations and Pareto frontier calculations at lower thresholds.

4. Independence assumption partially addressed. The Gaussian copula dependence model reduces the panel-level FP rate from 93.4% (independence) to 86.5% (copula-corrected), a 7.4% relative improvement. However, the 28 pairwise correlations are drawn from published studies in heterogeneous populations; patient-level paired biomarker measurements in a primary care chest pain cohort are needed for definitive calibration. The copula assumes a Gaussian dependence structure; tail dependence (relevant for extreme biomarker values) may differ. The corrected FP rate remains high (86.5%), confirming that dependence modelling mitigates but does not resolve the multi-analyte false-positive problem.

5. Serial testing extension. I extend the static framework with a time-dependent kinetics model incorporating ESC 0/1-hour and 0/3-hour serial algorithms. Published algorithm-level performance (Reichlin 2009: sensitivity 98.5% at 0/1h; Mueller 2019: 99.3% at 0/3h) is used as an override for ACS, while other pathologies are modelled via pathology-specific presentation-to-measurement delay ratios. This serial extension is exploratory—it uses published aggregate performance rather than patient-level delta calculations, and the HEAR score (without Troponin) distribution used for risk stratification has not been prospectively validated in Dutch primary care (Van den Bulk 2024).

6. Cost estimates and penalty weights. Costs are approximate estimates from device manufacturers and costing studies [66, 24]; actual costs vary by device and healthcare system. However, the weight sensitivity analysis proves that cost assumptions are irrelevant at $\tau = 0.90$.

7. Pneumothorax. Correctly identified as uncoverable by blood-based biomarkers at any threshold. This limitation reflects the biology, not the framework.

8. SISTER ACT score. The proposed score is a simulation-derived research proposal, not a validated CDR. Component weights and tier thresholds are theory-driven. The 93% e-stethoscope sensitivity for PTX derives from controlled studies [42, 41]; real-world performance in a noisy GP consulting room may differ. Prospective validation is essential before adoption.

9. Health-economic model limitations. The decision-analytic model uses a single-cycle decision tree rather than a Markov model capturing long-term downstream costs and outcomes. QALY losses per missed diagnosis are point estimates from clinical literature; individual variation is substantial. Cost inputs reflect 2024 Dutch NZa tariffs and are not generalisable to other healthcare systems without recalibration. The probabilistic sensitivity analysis varies all parameters simultaneously but assumes parametric distributions (Beta for sensitivities, Gamma for costs and QALYs) that may not capture the true uncertainty structure. The 100% cost-effectiveness probability reflects the strength of the dominance finding but should not be interpreted as certainty of real-world cost-effectiveness.

10. Sequential testing assumptions. The Bayesian sequential protocol assumes that biomarker test results are conditionally independent given disease status—a simplification, as the

copula analysis demonstrates non-trivial correlations. The early termination thresholds (discharge < 0.02, referral > 0.10) are chosen to prioritise safety but have not been calibrated against clinical outcome data. The information-gain ordering is optimal in an expected-value sense but may not be optimal for every individual patient.

Conclusion

Set-cover optimisation identifies a four-test POC panel (hs-cTnI, D-dimer, NT-proBNP, CRP) that rules out five of six life-threatening thoracic pathologies at $\tau = 0.90$, within 15 minutes, from a fingerprick sample. This solution is uniquely determined by the coverage constraints. Per-pathology NPV exceeds 0.99 for all covered pathologies. The panel-level false-positive rate of 93.4% under independence is reduced to 86.5% by Gaussian copula dependence modelling and further to 66.5% by quantitative likelihood-ratio interpretation—a combined 29% relative reduction. Pathology-directed management (ED referral only for ACS, PE, aortic dissection; outpatient workup for pericarditis and heart failure) reduces the ED referral rate to 63.0%, and the combination with HEAR pre-stratification achieves 370.5 unnecessary ED referrals per 1,000 patients (vs. 476.8 under binary interpretation, −22%). Bayesian sequential testing further mitigates operational impact, reducing the mean number of tests from 4 to 2.6 per patient (35% reduction) while preserving 100% sensitivity.

Health-economic analysis demonstrates that both the optimal panel and the SISTER ACT score dominate current care (HEART + hs-cTnI only): they are simultaneously cheaper and more effective. The optimal panel saves €140 per patient (209 additional diagnoses per 10,000 patients); SISTER ACT saves €618 per patient (245 additional diagnoses), with probabilistic sensitivity analysis confirming cost-effectiveness in 100% of 2,000 Monte Carlo iterations. Extrapolated nationally, SISTER ACT adoption could avert 3,670 missed serious pathology diagnoses and save €92.6M annually across the Dutch primary care system.

Expanding the candidate pool from 8 to 12 biomarkers (adding sST2, GDF-15, galectin-3, MR-proADM) confirms that the current panel is uniquely optimal at $\tau = 0.90$ but reveals alternative compositions at relaxed thresholds, demonstrating the framework's scalability. The ESC 0/1-hour serial troponin algorithm recovers ACS coverage at the GP measurement timepoint, and HEAR score stratification provides principled risk-based triage. Finally, the SISTER ACT score integrates the biomarker panel with an AI estethoscope, closing the pneumothorax coverage gap and achieving 6/6 pathology coverage (100%) at a marginal cost of €0.64 per patient—97.7% sensitivity at 70.0% specificity with a 38.2% referral rate.

The framework's value lies in making all trade-offs *transparent, quantifiable, and health-economically justified*, providing a reproducible template for multi-pathology panel design that bridges the gap between diagnostic accuracy studies and clinical deployment decisions.

Data availability

No new patient data were collected. All inputs derive from published diagnostic accuracy studies (25/48 entries) or expert estimates/indirect evidence (23/48) for off-target combinations. Complete code, coverage matrix, and results: `https://github.com/royerz2/PanDiagnosticCardiology`.

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Supplementary Tables

Table 10: Coverage matrix: published pooled sensitivity of each biomarker for each pathology. Bold entries exceed $\tau = 0.90$.

Pathology	hs-cTnI	D-dimer	NT-proBNP	CRP	Copeptin	H-FABP	Myoglobin	PCT
ACS	**0.95**	0.38	0.60	0.50	0.85	0.84	0.75	0.15
PE	0.45	**0.95**	0.60	0.65	0.55	0.48	0.30	0.35
Aortic Dissection	0.28	**0.97**	0.45	0.72	0.68	0.25	0.30	0.20
Pericarditis/Myo	0.82	0.35	0.70	**0.92**	0.55	0.68	0.58	0.40
Pneumothorax	0.10	0.15	0.20	0.15	0.30	0.05	0.08	0.10
Acute HF	0.68	0.60	**0.95**	0.55	0.78	0.55	0.40	0.30

Table 11: Full provenance for all 48 sensitivity values. CI = 95% confidence interval. †Expert estimate: no published meta-analysis for this biomarker–pathology combination.

Pathology	Biomarker	Sens	CI	Spec	Source
ACS	hs-cTnI	0.95	0.93–0.97	0.54	Lipinski 2015 [11]
ACS	D-dimer	0.38	0.28–0.49	0.45	Expert estimate†; non-specific
ACS	NT-proBNP	0.60	0.52–0.68	0.60	Expert estimate†; LV dysfunction
ACS	CRP	0.50	0.40–0.60	0.40	Expert estimate†; non-specific
ACS	Copeptin	0.85	0.79–0.90	0.75	Keller 2011 [19]; Raskovalova 2014 [20]
ACS	H-FABP	0.84	0.78–0.89	0.78	Bruins Slot 2010 [21]
ACS	Myoglobin	0.75	0.65–0.83	0.50	Lipinski 2015 [11]
ACS	PCT	0.15	0.08–0.24	0.85	Bektas 2011 [67]
PE	hs-cTnI	0.45	0.35–0.56	0.60	Becattini [68]
PE	D-dimer	0.95	0.93–0.97	0.42	Geersing 2012 [12]
PE	NT-proBNP	0.60	0.49–0.70	0.55	Klok [69]
PE	CRP	0.65	0.53–0.76	0.35	Abul [70]
PE	Copeptin	0.55	0.40–0.69	0.70	Hellenkamp [71]
PE	H-FABP	0.48	0.35–0.61	0.75	Puls [72]
PE	Myoglobin	0.30	0.18–0.44	0.60	Expert estimate†
PE	PCT	0.35	0.22–0.50	0.80	Köktürk [73]
AoD	hs-cTnI	0.28	0.18–0.40	0.65	Vrsalovic [74]
AoD	D-dimer	0.97	0.94–0.99	0.47	Nazerian [13]; Ren [14]; Chen [15]
AoD	NT-proBNP	0.45	0.32–0.59	0.55	Expert estimate†; limited data
AoD	CRP	0.72	0.58–0.83	0.30	Schillinger [75]
AoD	Copeptin	0.68	0.52–0.81	0.60	Morello [76]
AoD	H-FABP	0.25	0.12–0.42	0.80	Expert estimate†
AoD	Myoglobin	0.30	0.16–0.47	0.65	Expert estimate†
AoD	PCT	0.20	0.09–0.36	0.75	Expert estimate†
Peri/Myo	hs-cTnI	0.82	0.73–0.89	0.50	Caforio 2013 [77]
Peri/Myo	D-dimer	0.35	0.22–0.50	0.55	Expert estimate†
Peri/Myo	NT-proBNP	0.70	0.58–0.80	0.50	Caforio [77]
Peri/Myo	CRP	0.92	0.86–0.96	0.38	Imazio 2011 [18]
Peri/Myo	Copeptin	0.55	0.39–0.70	0.65	Expert estimate†

(continued)

Pathology	Biomarker	Sens	CI	Spec	Source
Peri/Myo	H-FABP	0.68	0.54–0.80	0.70	Cui [78]
Peri/Myo	Myoglobin	0.58	0.43–0.72	0.55	Expert estimate†
Peri/Myo	PCT	0.40	0.26–0.55	0.72	Bektas 2011 [67]
PTX	hs-cTnI	0.10	0.03–0.22	0.90	Expert estimate†; case reports
PTX	D-dimer	0.15	0.05–0.30	0.80	Expert estimate†
PTX	NT-proBNP	0.20	0.08–0.38	0.85	Expert estimate†; case series
PTX	CRP	0.15	0.05–0.30	0.70	Expert estimate†
PTX	Copeptin	0.30	0.14–0.50	0.65	Expert estimate†; stress response
PTX	H-FABP	0.05	0.01–0.18	0.90	Expert estimate†
PTX	Myoglobin	0.08	0.02–0.22	0.85	Expert estimate†
PTX	PCT	0.10	0.03–0.25	0.88	Expert estimate†
AHF	hs-cTnI	0.68	0.58–0.77	0.55	Felker [79]
AHF	D-dimer	0.60	0.48–0.71	0.40	Zorlu [80]
AHF	NT-proBNP	0.95	0.93–0.97	0.76	Maisel 2002 [16]; Bansal 2025 [17]
AHF	CRP	0.55	0.42–0.67	0.40	Windram 2007 [81] (prognostic)
AHF	Copeptin	0.78	0.67–0.87	0.55	Demissei 2017 [82]; Peacock 2011 [83]
AHF	H-FABP	0.55	0.42–0.67	0.70	Niizeki [84]
AHF	Myoglobin	0.40	0.28–0.53	0.60	Expert estimate†
AHF	PCT	0.30	0.18–0.44	0.80	Expert estimate†; PCT modest in HF

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